

PROMPT: 68-year-old male with progressive **dyspnea**, **orthopnea**, and leg edema. Echo shows EF **35%**, consistent with **heart failure**. ... mild pulmonary congestion; elevated creatinine suggestive of cardiorenal syndrome.

Task: Predict ICD-10 codes from a large set (10000+ labels).

I50.22 - Chronic systolic HF	J18.9 - Pneumonia	I42.9 - Cardiomyopathy
N18.3 - CKD (stage 3)	I25.10 - Coronary artery disease	N17.9 - Acute kidney failure
J44.9 - COPD	... +9000 unrelated codes	

semantic anchors (from text + candidate labels)

dyspnea **orthopnea** **EF 35%** **heart failure** **cardiorenal**

Early Chain-of-Thought Reasoning

Classic **heart failure** signs: **dyspnea**, **orthopnea**, edema. Echo: EF **35%** → systolic dysfunction. Consistent with chronic systolic HF (I50.22). Elevated creatinine → cardiorenal syndrome.

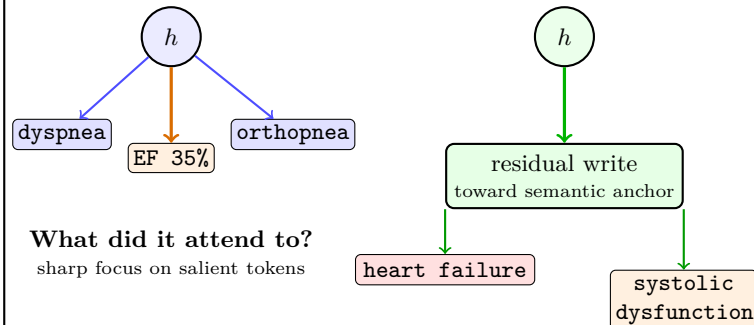
Strong semantic signals identified early

heart failure **dyspnea** **EF 35%** **orthopnea**

Mechanistic Question—Phase 1

When an early CoT token latches onto a strong semantic signal, what is an attention head doing internally?

early CoT token: **"heart failure"**



What did it attend to?
sharp focus on salient tokens

heart failure

systolic dysfunction

What did it write?
a semantic update in the residual stream